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MaskWarp: Visuo-Haptic Illusions in Mixed Reality using Real-time Video Inpainting

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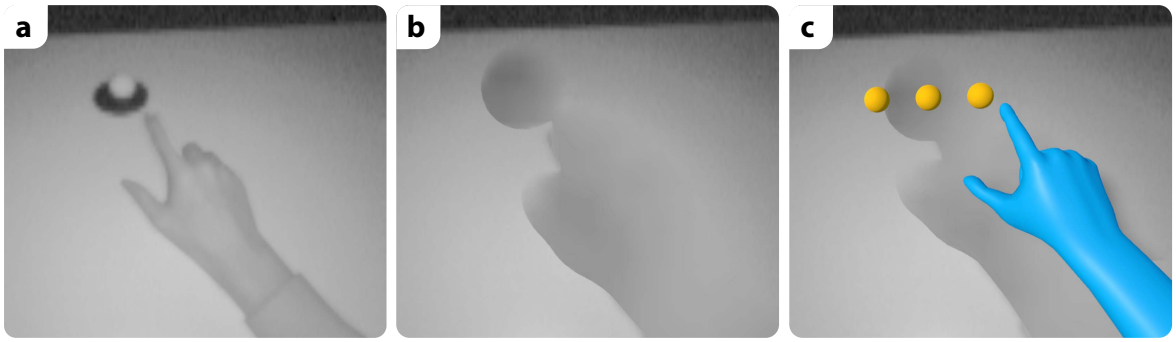

Figure 1: a) The user can see their real hand, objects and environment in video see-through mixed reality. b) MaskWarp uses video inpainting to remove the target (in this case a ball) and hand from the video. c) The virtual hand position can now be warped for haptic retargeting reusing one physical ball for three virtual balls. A Leap Motion is used for hand tracking and video see-through.

Abstract

We present MaskWarp, a novel approach to enable haptic retargeting and other visuo-haptic illusions in video see-through (VST) mixed reality (MR). Virtual reality provides a unique ability to shift where users perceive their virtual hands. However, in VST MR, users can see their real hands through a video feed, breaking the illusion. MaskWarp uses real-time video inpainting to remove the physical hand from the video, which is then replaced with a virtual hand. The virtual hand position can then be warped for visuo-haptic illusions. We use haptic retargeting to demonstrate this technique in an engaging cups and ball game.

Index Terms: Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality Computer graphics—Graphics systems and interfaces—Perception

1 Introduction

Immersive virtual reality (VR) provides a unique ability to warp where users perceive their virtual hands. Visuo-haptic illusions like hand redirection and haptic retargeting [1] rely on the dominance of vision over the other senses, supported in VR by the user's inability to see their real hand. In video see-through (VST) mixed reality (MR), the user sees the real world and their real hand through a stereoscopic video feed as shown in Fig. 1a. Virtual content is drawn over the video, enabling users to engage with the real and virtual worlds. This virtual content can fully occlude the user's view of the real world. The video can also be modified to control what the user sees using filters and computer vision methods.

Previous work has explored visuo-haptic shape change illusions in MR by blocking the real hand with virtual objects, obscuring the entire desk area around the physical object [5]. MaskWarp uses real-time video in-painting to hide just the user's real hand and passive haptic proxy. The hand and proxy are removed from the camera feed, and the resulting hole is filled based on information from surrounding pixels, shown in Fig. 1b. This technique of removing information from the visual environment and replacing it with virtual information is also known as diminished reality [4].

With their real hand concealed, the user is given a virtual hand for interacting with virtual information. Fig. 1c shows how this virtual hand can be warped to create visuo-haptic illusions like hand redirection and haptic retargeting. We present the implementation of our initial prototype and an entertaining demonstration of haptic retargeting in MR using a cup and ball tracking game. We then outline future research directions for this concept to develop the idea further and understand the operational limits.

2 MaskWarp Implementation

The prototype is implemented in Unity 2021.3.27f1 using an Oculus Rift S with Leap Motion hand-tracking. The Leap Motion was selected for both the hand tracking and the stereoscopic camera; however, it only provides black-and-white images. Using the same camera for the video and hand tracking ensures consistency between the tracked hand pose and the hand in the video. The Leap Motion also supports forearm tracking, enabling the removal of both the hand and forearm from the video. If precise alignment between hand tracking and video can be achieved, our implementation is compatible with other stereoscopic camera systems.

The first step in the MaskWarp pipeline shown in Fig. 2 is to produce the inpainting mask. The tracked hand and passive haptic model are rendered in white on a black background. A Gaussian blur filter is applied to the mask, followed by a threshold operation to saturate the blurred areas expanding the masked areas. The mask is composited with the current frame from the VST camera, providing a single image where masked areas are transparent.

We follow a pyramid-based inpainting approach [2]. While it is quite limited in realism, it is suitable for parallel implementation and sufficient for this initial prototype. First, the composited image is downsampled, generating successively smaller mipmaps. Every 2x2 region of pixels is downsampled to one by averaging the colours but excluding masked pixels. Regions with only mask pixels remain

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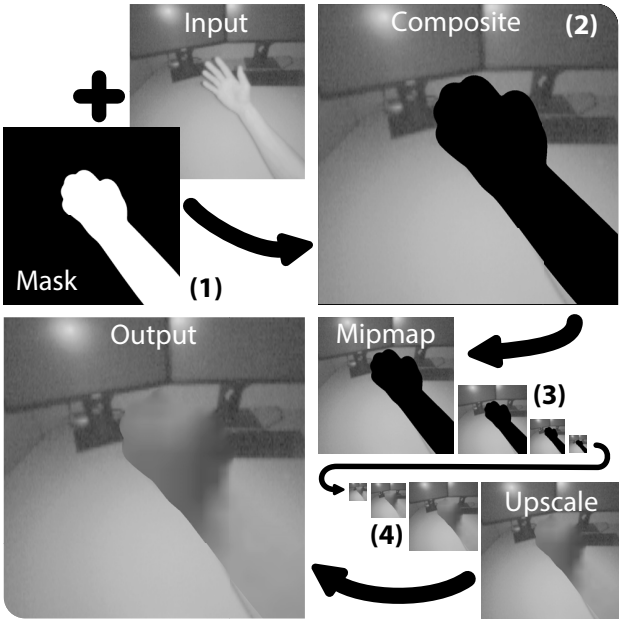

Figure 2: The inpainting pipeline involves four key steps: 1) creating the mask, 2) compositing the mask and video, 3) mipmapping the non-mask pixels, and 4) upscaling while filling mask pixels from smaller maps to generate the output. Here the technique is shown in a typical desktop environment to show how the masked area is filled with surrounding information.

masked in the resulting image until no masked pixels remain. Then, working from the second smallest frame to the largest, the frames are upscaled, filling the remaining masked areas with information from the previous smaller frame. This approach fills the masked areas using lower fidelity information from the surrounding pixels.

Without direct visual knowledge of the hand position, the user can instead rely on the virtual hand, allowing for hand redirection and haptic retargeting. Inpainting is only enabled while the user's hand is redirected enabling them to easily interact with objects and people in the real world.

3 Demonstration

We demonstrate MaskWarp with a cups and ball game shown in Fig. 3. In MR, the participant will see the demonstrator place a ball under a single physical cup. Then using MaskWarp the cup is replaced with a virtual version with two additional replicas on the left and right. The cups shuffle around, and the participant tries to track which cup the ball is under. As they reach for their selected cup, haptic retargeting redirects them to grab and lift the single physical cup. The real ball is also masked so the user cannot see it when they lift the cup. When the participant reveals the hidden object, the game is complete.

We use a hand pointer with a 0.3-second dwell time to let the participant select which cup they want to lift. Shape Aware Haptic Retargeting [3] provides accurate warping for interactions with the whole hand using a cylinder to approximate the cup shape and fingertip positions on the hand.

With this interactive experience, we aim to show several opportunities the combination of haptic retargeting, and MR provide: 1) integrating real-world elements (placing a real ball under a real cup), 2) retaining the haptic feedback of real objects for scalable and dynamic virtual experiences, and 3) satisfying visual effects. This provides a unique experience by blending the real and virtual worlds

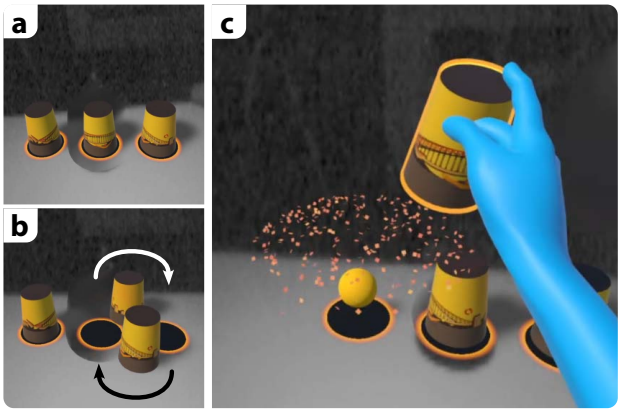

Figure 3: The cup and ball tracking game demo. (a) The initial three cups with a ball under the centre cup aligned with a physical ball and cup. (b) The cups randomly swap locations. (c) The participant lifts the correct cup, revealing the hidden ball.

where virtual elements can be physically grabbed and interacted with. This demonstration is both entertaining and visually engaging. It demonstrates MaskWarp using a relatively unique concept in haptic retargeting, which most people have not yet experienced.

4 Conclusion and Future Work

Although the hand is removed from the video, MaskWarp still gives the user additional cues about their real hand position, like the blurry masked region and shadows. Ideally, hand redirection should remain undetectable to the user. In future work, we will investigate the detection thresholds for hand redirection in MR using MaskWarp. The inpainting mask is generated using the hand-tracking pose, resulting in inaccuracies and requiring an oversized mask to remove the hand. Hand and object detection could be applied directly to the image frames to find a more precise mask and reduce the inpainted area. We also intend to explore advanced real-time video inpainting methods and colour stereoscopic cameras for more realistic results.

We believe this initial prototype and demonstration will provide a foundation for future study of visuo-haptic illusions in MR. Furthermore, we hope that extending hand redirection and haptic retargeting beyond VR will lead to new and exciting applications.

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